

PASSWORD GENERATOR [GitHub](#)

Project Type: A Python-based application that generates secure and customizable passwords.

Programming Language: Python

Internship: Python Developer Internship

Organization: Auspify Technologies

1. Description

The Password Generator is a Python console application that creates random passwords based on the length entered by the user.

The password contains letters, numbers, and special characters. The application also validates user input and allows the user to generate multiple passwords.

2. Objective

The main objective of this project is to:

- Generate secure random passwords.
 - Allow users to choose the password length.
 - Include letters, numbers, and symbols.
 - Validate user input.
 - Practice basic Python programming concepts.
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3. Features

- Secure Password Generation
- Custom Password Length
- Uppercase & Lowercase Letters
- Numbers
- Special Characters
- Input Validation
- Generate Multiple Passwords
- Error Handling

The program ensures that every password has at least one letter, one number, and one symbol.

4. Technologies Used

Technology	Purpose
Python	Develop the application
String Module	Provide letters, numbers and symbols
Secrets Module	Secure random character generation
Console	User interaction

5. How to Run

Follow these simple steps to run the Password Generator:

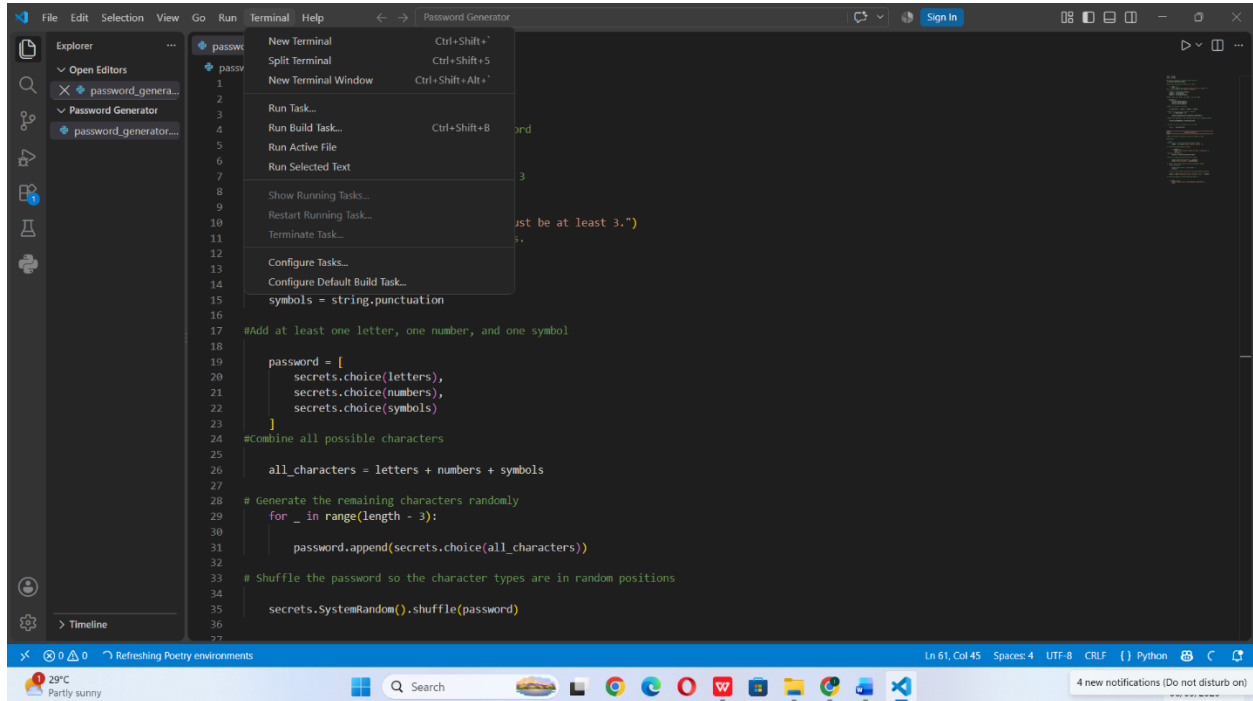
Step 1: Install Python on your computer.

Step 2: Save the program as:

password_generator.py

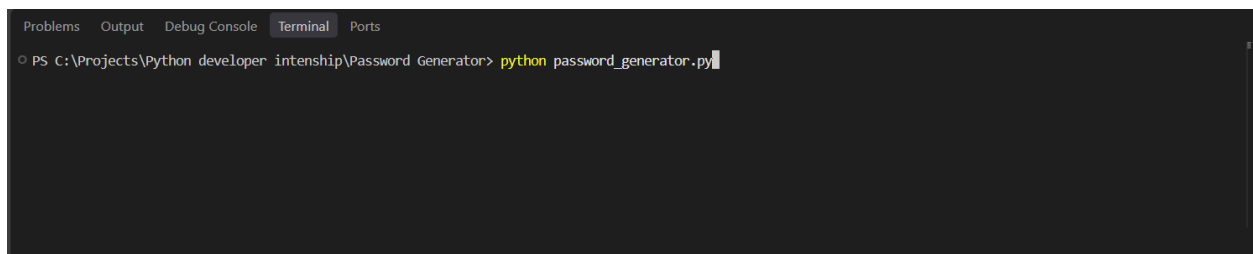
Step 3: Open the project folder where the Python file is saved.

Step 4: Open Command Prompt / Terminal in that folder-> New terminal



Step 5: Run the program using:

python password_generator.py



Step 6: Enter the required password length when prompted.

Enter password length: 12

```
Problems Output Debug Console Terminal Ports
PS C:\Projects\Python developer intenship\Password Generator> python password_generator.py
=====
                        PASSWORD GENERATOR
=====

Enter password length: 12

Generated Password: "f9IR\tF#Lt3
Password Length: 12
```

Step 8: Enter **y** to generate another password or **n** to exit.

Generate another password? (y/n): n

```
Problems Output Debug Console Terminal Ports
PS C:\Projects\Python developer intenship\Password Generator> python password_generator.py
Enter password length: 12

Generated Password: "f9IR\tF#Lt3
Password Length: 12

Generate another password? (y/n): y
Enter password length: 
```

```
Problems Output Debug Console Terminal Ports
PS C:\Projects\Python developer intenship\Password Generator> python password_generator.py
Enter password length: 12

Generated Password: (!Q/C3eLD'2b
Password Length: 12

Generate another password? (y/n): n

Thank you for using Password Generator!
PS C:\Projects\Python developer intenship\Password Generator> 
```

Result: The Password Generator runs successfully and generates a secure password based on the user's selected length.